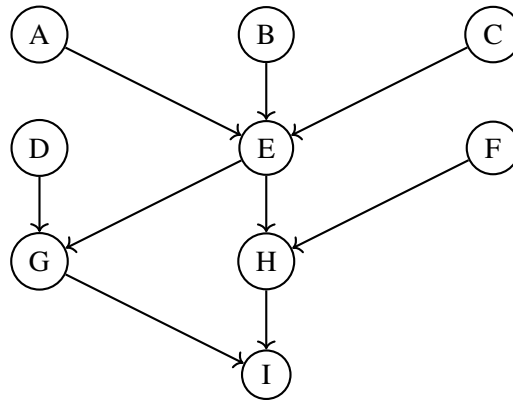


50.007 Machine Learning, Spring 2025 Homework 4

Due on April 21, 2025, 11.59 pm

In this homework, we explore Bayesian Networks. The network structure is shown below. All nodes take two values: $\{1, 2\}$.



Question 1 [15 points]

- (a) Without knowing the actual value of any node, are node A and node F independent of each other?
- (b) What if we know the value of node C and node I ?
- (c) Further, analyze the cases when only node C is observed and when only node I is observed. Provide a justification using the Bayes' ball algorithm.

Answer:

- (a) The path

$$A \rightarrow E \rightarrow H \leftarrow F$$

has H as a collider which is unobserved (and none of its descendants are observed), so this path is blocked. Hence A and F are marginally independent.

- (b) If we observe C and I , consider the path

$$A \rightarrow E \rightarrow G \rightarrow I \leftarrow H \leftarrow F.$$

Here I is a collider; conditioning on I opens this path. Therefore A and F become dependent given C, I .

- (c)

- Observing only C does not open any collider, so $A \perp F \mid C$.
- Observing only I opens the collider at I , so A and F become dependent given I .

Question 2 [15 points]

(a) Derive the formula for the *effective number of parameters* for a node i given its state space r_i and the state spaces of its parents $\prod_{j \in \text{pa}_i} r_j$. (b) Compute the effective number of parameters needed for this Bayesian network when all nodes are binary (i.e., taking values $\{1, 2\}$). (c) What would be the number of parameters if node D and node F can take 5 different values: $\{1, 2, 3, 4, 5\}$, and all other nodes can only take 3 different values: $\{1, 3, 5\}$? Explain how the increased state space affects the model complexity.

Answer:

(a) **General formula** For node i with r_i outcomes and parents whose joint state-space has size

$$q_i = \prod_{j \in \text{pa}_i} r_j,$$

its CPT has q_i rows and r_i columns, with one degree of freedom lost per row (they sum to 1). Hence the number of free parameters is

$$(r_i - 1) q_i = (r_i - 1) \prod_{j \in \text{pa}_i} r_j.$$

(b) **All-binary case** Here $r_i = 2$ for every node:

- A, B, C, D, F (no parents): each has $(2 - 1) = 1$ parameter, total $5 \times 1 = 5$.
- E (three binary parents): $q_E = 2^3 = 8$, free parameters $8 \times (2 - 1) = 8$.
- G, H, I (each with two binary parents): $q = 2^2 = 4$, parameters $4 \times 1 = 4$ each, total $3 \times 4 = 12$.

$$5 + 8 + 12 = 25.$$

(c) **Mixed state-spaces** Let $r_D = r_F = 5$, and all others $r_i = 3$ (values $\{1, 3, 5\}$):

- A, B, C : no parents, each $(3 - 1) = 2$, total $3 \times 2 = 6$.
- D, F : no parents, each $(5 - 1) = 4$, total $2 \times 4 = 8$.
- E (parents A, B, C each 3-valued): $q_E = 3^3 = 27$, parameters $27 \times (3 - 1) = 54$.
- G (D 5-valued, E 3-valued): $q_G = 5 \times 3 = 15$, parameters $15 \times (3 - 1) = 30$.
- H (E 3-valued, F 5-valued): similarly 30.
- I (G, H each 3-valued): $q_I = 3^2 = 9$, parameters $9 \times 2 = 18$.

$$6 + 8 + 54 + 30 + 30 + 18 = 146.$$

Question 3 [15 points]

(a) Given the probability tables below, compute $P(E = 2 \mid C = 1)$, clearly writing down all necessary steps. (b) Now compute $P(E = 2 \mid C = 1, B = 2)$. Discuss how the additional observation affects the inference, and explain which conditional independencies simplify the calculation.

Answer:

(a)

$$P(E = 2 \mid C = 1) = \frac{P(C = 1, E = 2)}{P(C = 1)} = \frac{P(C = 1) \sum_{A,B} P(A)P(B)P(E = 2 \mid A, B, C = 1)}{0.2}.$$

Compute the sum over (A, B) :

$$\begin{aligned} &0.2 \cdot 0.7 \cdot 0.9 + 0.2 \cdot 0.3 \cdot 0.5 + 0.8 \cdot 0.7 \cdot 0.1 + 0.8 \cdot 0.3 \cdot 0.5 \\ &= 0.126 + 0.03 + 0.056 + 0.12 = 0.332. \end{aligned}$$

Thus,

$$P(C = 1, E = 2) = 0.2 \times 0.332 = 0.0664, \quad P(E = 2 \mid C = 1) = 0.0664/0.2 = 0.332.$$

(b)

$$P(E = 2 \mid C = 1, B = 2) = \sum_A P(A)P(E = 2 \mid A, B = 2, C = 1) = 0.2 \cdot 0.5 + 0.8 \cdot 0.5 = 0.5.$$

Given B , we no longer sum over B , and since both A cases yield the same conditional probability, the result is 0.5.

Question 4 [10 points]

Assume we have the following 12 complete observations. Find a way to estimate the probability table for node A and for node H .

Answer:

Estimate for Node A : Out of 12 samples, $A = 1$ appears 7 times and $A = 2$ appears 5 times:

$$P(A = 1) = \frac{7}{12}, \quad P(A = 2) = \frac{5}{12}.$$

Estimate for Node H (parents E, F): Count occurrences for each (E, F) :

- $(E, F) = (1, 1)$: 1 sample, $H = 1$, so $P(H = 1 \mid 1, 1) = 1$, $P(H = 2 \mid 1, 1) = 0$.
- $(E, F) = (1, 2)$: 3 samples, $H = 1$ twice, $H = 2$ once, so $P(H = 1) = 2/3$, $P(H = 2) = 1/3$.
- $(E, F) = (2, 1)$: 6 samples, $H = 1$ four times, $H = 2$ twice, so $P(H = 1) = 2/3$, $P(H = 2) = 1/3$.
- $(E, F) = (2, 2)$: 2 samples, one each, so $P(H = 1) = 1/2$, $P(H = 2) = 1/2$.